

Lightweight Deep Learning for Agriculture: A Computer Vision Approach to Plant Disease Detection

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Abstract

This paper intends to meet the important demand for early and precise detection of plant diseases, which has a substantial influence on agricultural output and sustainability. The paper's goal is to create a system that can classify plant diseases from photos using computer vision and deep learning. The goal is to properly preprocess plant photos and utilize advanced neural networks, including custom models and the VGG16 pretrained model, to achieve reliable and scalable disease detection. This contribution lays the groundwork for automated disease detection, allowing for more prompt intervention and better agricultural outcomes.

Keywords: Deep Learning, Computer Vision, Plant Disease Detection, Convolutional Neural Networks (CNNs), VGG16, Agricultural Productivity, Image Preprocessing.

1. Related Work

An advanced deep convolutional neural network (CNN) was proposed in this study (Geetharamani & Arun Pandian, 2019) for diagnosing plant leaf diseases using a dataset containing 39 different classes of plant leaves, including both healthy and diseased samples. In contrast to conventional machine learning models, where feature extraction is typically performed manually, deep CNN utilized an end-to-end learning approach, enhancing classification capabilities. The authors demonstrated that incorporating data augmentation techniques—such as flipping, gamma correction, noise injection, Principal Component Analysis (PCA) color augmentation, rotation, and scaling—helped increase dataset size and improve model generalization. The final model, M4-Both, was evaluated, achieving an average accuracy of 96.46%, surpassing other transfer learning techniques.

In another study (Rao et al., 2021), transfer learning was applied using AlexNet for plant disease detection, specifically in grape and mango leaves. The model was trained on a dataset containing 8,438 images, including 7,222 images from the PlantVillage dataset and 1,266 newly collected mango leaf images. The developed model achieved an accuracy of 99.03% for grape leaves and 89% for mango leaves. The system was implemented using MATLAB and integrated into the "JITCROPFIX" mobile application, allowing farmers to diagnose diseases and minimize economic losses. The primary contribution of this work is demonstrating the potential of deploying pre-trained CNN models to address agricultural challenges, particularly in regions such as India, where limited

technological infrastructure hinders rapid disease identification. However, key limitations were acknowledged, including data scarcity across different geographic locations and lighting conditions, as well as challenges in detecting minor deformities at early stages. Future research directions involve extending the model to detect additional plant diseases, implementing a recommendation system for disease management, and integrating the system with self-driving drones for field surveillance. These advancements highlight how artificial intelligence (AI) is transforming precision farming by improving cost efficiency and scalability in disease prevention.

This paper introduces a deep learning-based object detection model designed for multi-class plant disease detection, with a particular focus on apple plants. The model employs an enhanced You Only Look Once version 4 (YOLOv4) configuration, pre-trained on the Common Objects in Context (COCO) dataset from GitHub and fine-tuned for improved computational efficiency. To address the challenge of detecting diseases at multiple scales and with high precision, the proposed model achieved a mean Average Precision (mAP) of 0.912, an F1-score of 0.959, and a frame rate of 56.9 frames per second (FPS) (Roy & Bahduri, 2021). This model is particularly effective in real-world agricultural environments, where many previous models struggle to detect small-scale infections and early-stage diseases. The system outperformed prior models, achieving a 9.05% higher precision and a 7.6% higher F1-score, demonstrating its effectiveness in real-time disease diagnosis. Additionally, the research suggests that the model can be applied to other agricultural domains, including disease detection in various crops, automated monitoring systems, and scalable precision farming solutions. The findings indicate that improved disease detection efficiency through this model will enhance disease prevention strategies, leading to significant economic and environmental benefits.

A recent study utilizing the same dataset evaluated various transfer learning models for classifying 38 plant disease classes. Among the tested architectures, DenseNet-121 outperformed ResNet-50, VGG16, and Inception V4, achieving a classification accuracy of 99.81% and an F1-score of 99.8% (Eunice et al., 2022). DenseNet-121's superior performance was attributed to its reduced number of trainable parameters and lower computational complexity, making it a promising model for extending classification to additional plant diseases.

The identification of diseases in plant crops, such as olive tree diseases, is essential for effective disease management, particularly in regions like Saudi Arabia (Ksibi, Ayadi,

Soufiene, Jamjoom, & Ullah, 2022). Traditional disease detection methods are often slow and unreliable, highlighting the significance of deep learning (DL) in improving accuracy and automation in disease diagnosis. A novel model, MobiRes-Net, was introduced by integrating ResNet50 and MobileNet through deep feature concatenation, enhancing disease detection in olive leaves. The proposed model, trained on images captured from an unmanned aerial vehicle (UAV)-mounted camera, achieved a 97.08% classification accuracy, outperforming its individual components, ResNet50 (94.86%) and MobileNet (95.63%). However, the MobiRes-Net model requires significant training time and relies on high-performance computing resources. Future improvements could involve fine-tuning the model for real-time implementation and integrating it with field-programmable gate arrays (FPGAs) to enhance computational efficiency. Given its strong performance in identifying diverse plant diseases, the model has the potential for broader applications in agricultural disease management.

Based on the findings of Bagga and Goyal (2023), deep learning (DL) models, particularly Convolutional Neural Networks (CNNs), have significantly advanced plant disease diagnosis, achieving accuracies ranging from 85% to over 90%. Model performance has been enhanced through preprocessing techniques such as image scaling, noise filtering, and contrast enhancement. Transfer learning with pre-trained models, including Residual Network (ResNet), Inception, and Visual Geometry Group (VGG) networks, has proven particularly effective, enabling high accuracy even with smaller datasets while minimizing the need for extensive training from scratch.

Image segmentation, commonly applied alongside CNN architectures, facilitates the identification of specific disease-affected regions in plants, thereby improving model accuracy. Various data augmentation strategies, including random rotations and brightness adjustments, are frequently utilized to enhance model generalization and reduce overfitting, particularly when working with limited data. Additionally, CNN architectures such as AlexNet, VGG, and Inception are widely implemented using deep learning frameworks like TensorFlow and Keras, allowing for efficient fine-tuning of models for specific plant diseases. These advancements highlight the effectiveness of CNNs in automating plant disease identification, offering improved accuracy and processing speed compared to traditional methods, especially when integrated with comprehensive data preprocessing and transfer learning techniques.

In a study conducted by Ma et al., a Deep Convolutional Neural Network (DCNN) was developed to diagnose

cucumber diseases based on visible symptoms. Disease symptoms were extracted from cucumber leaf images captured in the field using an image segmentation algorithm, which effectively distinguished diseased regions from background elements. The study utilized a dataset containing four specific diseases: anthracnose, downy mildew, powdery mildew, and target leaf spots. Data augmentation techniques were applied to improve model robustness.

Quantitative evaluation demonstrated that the DCNN achieved a recognition accuracy of 93.4% on an imbalanced dataset and 92.2% on a balanced dataset. Comparative analysis revealed that AlexNet outperformed the DCNN due to its richer feature representations. However, both AlexNet and the DCNN demonstrated superior classification performance compared to traditional machine learning classifiers.

2. Technical Approach

This section discusses the various approaches applied for image preprocessing to enhance classification performance by improving data quality and enabling the model to process images more effectively. Additionally, different modeling approaches are explored and evaluated to determine the most suitable one for plant disease classification.

The following figures illustrate the distribution of classes within the dataset, along with the number of images associated with each class.

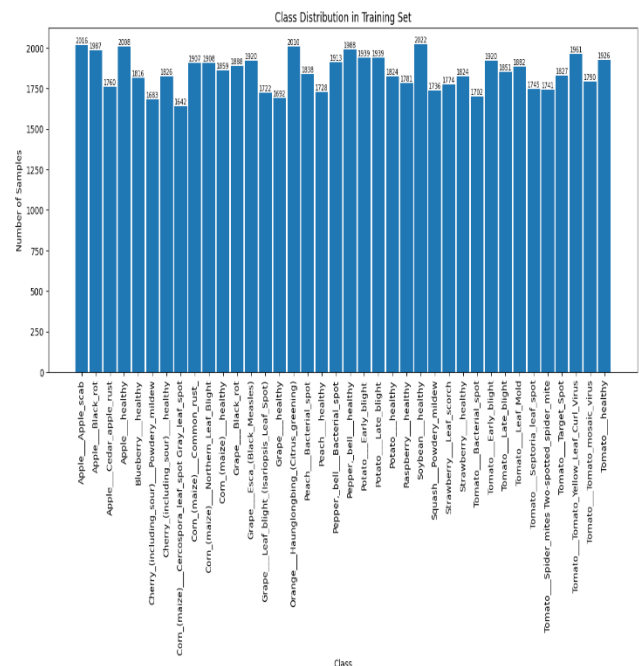


Figure 1 - Training Set

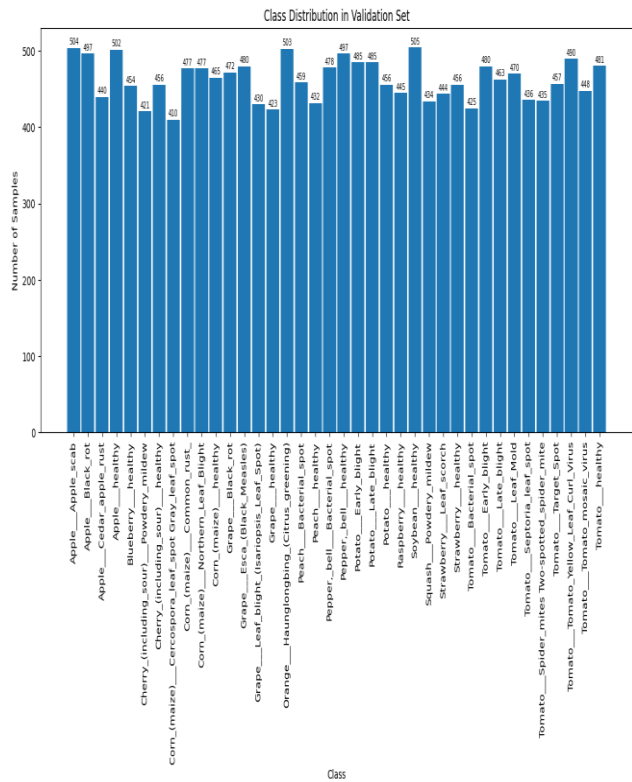


Figure 2 – Validation Set

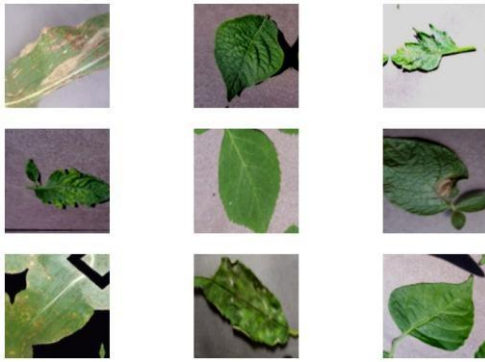


Figure 3 – A sample of the images we are using

2.1 Preprocessing techniques

2.1.1 Image Transformation

The first image preprocessing approach involved designing an algorithm to process images and detect brownish areas, which represent infected regions in the dataset. The Canny edge detector was applied to overlay detected edges onto the original image.

Initially, each image was loaded and converted from the default Blue-Green-Red (BGR) color format to Red-Green-

Blue (RGB) for proper visualization. The image was then transformed into the Hue, Saturation, Value (HSV) color space, which is more effective for color-based segmentation.

A specific range was defined for detecting brownish colors in HSV, with a lower bound of [10, 50, 50] and an upper bound of [30, 255, 200]. Using these bounds, a binary mask was generated to highlight all pixels within the specified brownish range. Morphological operations, specifically a closing operation with a rectangular kernel, were applied to remove noise and fill small gaps in the mask.

The identified brown regions were then highlighted in the original image by performing a bitwise AND operation between the image and the mask. Simultaneously, edges were detected in the grayscale version of the image using the Canny edge detector, which identifies boundaries with significant intensity variations, indicating infected plant areas.

To visually integrate the results, the detected edges were converted to RGB format and overlaid onto the brown-highlighted image using a weighted sum. This process generated a composite output where both brownish regions and edges were distinctly visible.

2.1.2 Equalization

Histogram equalization was applied as the second preprocessing method to enhance image contrast, ensuring that key features became more distinguishable. This technique improved the quality and uniformity of input data, enabling the model to better recognize patterns while reducing sensitivity to variations in lighting or exposure across different images. By standardizing brightness and contrast, the model focused on meaningful features rather than being affected by inconsistencies in the data.

The implemented algorithm first loaded the image in BGR format, which was then converted to RGB format for compatibility with standard visualization tools. The image was then split into its three color channels (Red, Green, and Blue), with histogram equalization applied separately to each channel. This technique redistributed brightness levels across the image, enhancing contrast by spreading out the intensity distribution of pixel values, thereby improving differentiation between light and dark regions.

After equalization, the three color channels were merged to reconstruct the processed image. The final enhanced image was converted back to RGB format, ensuring optimal visualization and maintaining consistency with the dataset.

2.1.3 Shadow Removal

Shadow removal was applied as a preprocessing technique to enhance image uniformity for improved processing and analysis. Eliminating shadows is essential in image preprocessing, as shadows can obscure critical details and introduce variability in the dataset.

The process begins by reading the image in Blue-Green-Red (BGR) format and converting it into the YCbCr color space. This transformation is performed because YCbCr separates brightness (Y) from chromatic information (Cb and Cr), facilitating shadow analysis based on intensity levels. A copy of the YCbCr image is then generated to serve as a binary mask, classifying shadow and non-shadow regions.

The mean and standard deviation of the Y channel (brightness) are computed to establish a threshold for shadow classification. Pixels with intensity below this threshold—defined as the mean minus a fraction of the standard deviation—are classified as shadow pixels and marked as white in the binary mask, while non-shadow pixels are marked as black.

To refine shadow detection and minimize misclassification, morphological erosion is applied to the binary mask. The algorithm then calculates the sum and count of pixel intensities separately for shadow and non-shadow regions. Using these values, the average brightness in both regions is determined, along with the intensity difference between them.

To normalize the shadowed areas, an adjustment is applied using the intensity difference and a ratio derived from the average brightness levels. This ensures that shadowed regions are adjusted to match the brightness of the illuminated areas, effectively removing shadows while preserving essential image details.

2.1.4 Intrinsic Image

The final image preprocessing approach focused on reducing the impact of lighting variations by separating object colors from illumination. This technique ensures that the model concentrates on the true features of the image, enhancing consistency and robustness in analysis.

The preprocessing method is based on Retinex theory, which decomposes an image into two components: reflectance, representing the true colors of objects, and illumination, capturing the lighting conditions. This decomposition enhances image features by minimizing the influence of varying lighting conditions, making it particularly useful for tasks such as object recognition and shadow removal.

The implemented algorithm processes each image by first loading it and converting it to Red-Green-Blue (RGB) format. The image is then normalized to the range $[0,1]$ and transformed into the logarithmic domain to simplify subsequent calculations. Illumination is estimated using a Gaussian blur, which smooths the image, while reflectance is computed by subtracting the smoothed illumination from the logarithmic image. Both components are then converted back to their original scale for further processing.

2.2 Modelling techniques

For modeling, two different approaches were evaluated: a pretrained Visual Geometry Group 16 (VGG16) model and a custom-designed Deep Convolutional Neural Network (CNN).

2.2.1 VGG16

VGG16, a Convolutional Neural Network (CNN) model, was utilized for plant disease classification. As a pretrained model on the ImageNet dataset, it has already learned to recognize a wide range of general image features, making it an effective choice for transfer learning.

VGG16 processes plant images by extracting meaningful features through its convolutional layers, which are well-suited for identifying patterns and textures in leaves and stems. These extracted features are then used to classify images into various disease categories. By leveraging its pretrained weights, the focus was placed on fine-tuning the model for plant disease classification, minimizing the need for extensive training from scratch.

2.2.2 CNN

The deep Convolutional Neural Network (CNN) is specifically designed for multi-class image classification tasks, with the capability to classify images into 38 distinct categories. The model processes input images with dimensions $128 \times 128 \times 3$, representing Red-Green-Blue (RGB) color channels, and is implemented using the TensorFlow Keras Sequential API. Its architecture consists of a series of convolutional blocks, followed by fully connected layers to ensure comprehensive feature extraction and accurate classification.

The convolutional architecture comprises five hierarchical blocks, each containing two convolutional layers with 3×3 filters. The number of filters increases progressively from 32 to 512 across the blocks, enabling the model to extract features of increasing complexity. The initial layers focus on learning basic patterns, such as edges and textures, while

deeper layers capture more intricate features. Max-pooling layers follow each block to reduce the spatial dimensions of the feature maps, optimizing computational efficiency and minimizing the risk of overfitting.

To further mitigate overfitting, dropout layers are incorporated at key stages of the network. A dropout rate of 0.25 is applied after the convolutional blocks, and a dropout rate of 0.4 is used before the dense layers. These layers deactivate a fraction of neurons during training, enhancing the model's generalization capabilities. The network concludes with a dense section, where extracted features are flattened into a single vector and passed through a fully connected layer with 1,500 units and Rectified Linear Unit (ReLU) activation. The final layer employs a softmax activation function to generate a probability distribution over the 38 target categories. This architecture demonstrates robustness and efficiency in multi-class classification tasks, providing an effective balance between feature extraction and computational demands.

3. Results

The initial step involved designing a data augmentation pipeline to enhance the diversity of training images and improve the model's ability to generalize. The implemented code applies random transformations, including flipping, rotation, zooming, and adjustments to brightness and contrast, to simulate real-world variations.

Additionally, pixel values are normalized to a range of $[0,1]$ to ensure numerical stability during training. These augmentations increase the model's robustness to variations in input data and enrich the training set by introducing additional variability, thereby improving generalization to real-world image conditions.

Following these image augmentation transformations, two modeling approaches were evaluated: the custom-designed Convolutional Neural Network (CNN), which achieved an accuracy of 97.35%, and the pretrained Visual Geometry Group 16 (VGG16) model, which resulted in an accuracy of 89.23%.

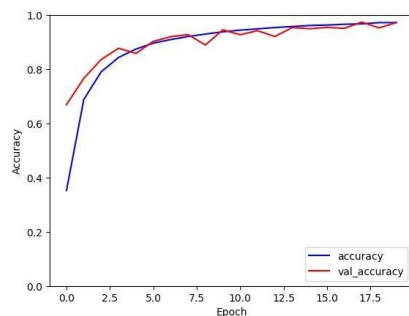


Figure 4 – Accuracy of the CNN model

Contrary to initial expectations, the results demonstrated greater robustness and accuracy when preprocessing techniques were omitted, and only image generalization transformations were applied. This outcome suggests that certain preprocessing methods may have introduced distortions or reduced the model's ability to generalize effectively.

3.1 Image Transformation

Below are the results of this technique.

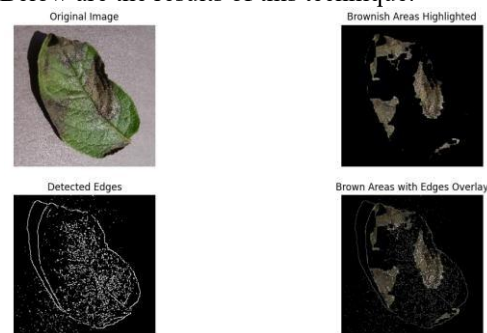


Figure 5 – Image Transformation Results

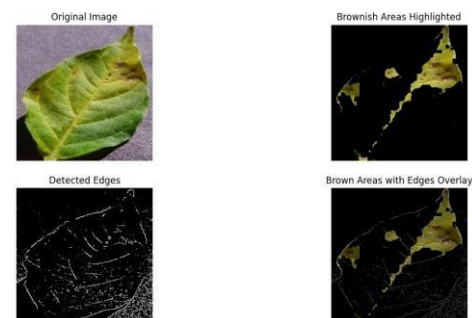


Figure 6 – Image Transformation Results

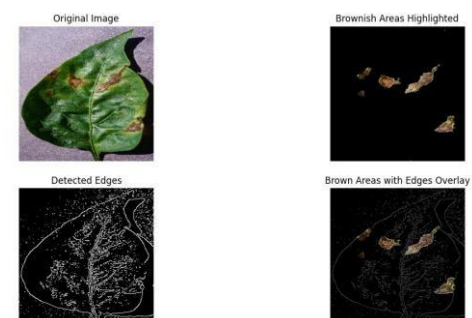


Figure 7 – Image Transformation Results

Although this technique demonstrated success across a wide range of images, certain corner cases presented challenges. In some instances, infected regions of the plants

did not exhibit a brownish color, leading to misclassification by the preprocessing approach. Additionally, in some cases, the disease manifested as cuts or missing portions of the plant, which were not effectively detected using this method. Furthermore, certain images contained brown elements in the background or on the plant itself, unrelated to the disease, which were incorrectly identified as infected areas.

These limitations introduced distortions in the model's learning process, ultimately leading to a reduction in classification accuracy. To address these issues, advanced segmentation techniques, such as semantic segmentation models or region-based methods, could be incorporated to enhance the detection of infected areas. These approaches would enable the model to focus on plant-specific features rather than relying solely on color-based preprocessing, thereby improving overall classification performance.

3.2 Equalization

Below are the results of this technique.



Figure 7 – Image Equalization Results



Figure 8 – Image Equalization Results



Figure 9 – Image Equalization Results

Although this approach enhanced image contrast, it negatively impacted model accuracy and led to overfitting,

likely due to color alterations across multiple regions of the image.

To mitigate this limitation, adaptive histogram equalization could be applied selectively or combined with other preprocessing techniques that preserve color consistency. This approach would enhance image quality while maintaining the model's ability to generalize effectively, preventing performance degradation.

3.3 Shadow Removal

Below are the results of this technique.



Figure 10 – Shadow Removal Results

This approach presented a significant challenge, as shadows were often present not only in the background but also on the plant itself. In such cases, shadow removal resulted in misleading and unreliable images, leading to inaccurate model predictions.

A potential solution to this issue is the incorporation of advanced shadow detection techniques capable of distinguishing between shadows on the plant and those in the background. This would ensure more reliable preprocessing while preserving the integrity of the input images, ultimately improving classification accuracy.

3.4 Intrinsic image

Below are the results of this technique.



Figure 11 – Intrinsic Image Results

This approach altered image contrast, yielding positive results for certain data samples but proving ineffective for the majority. Attempts to improve performance through

fine-tuning did not yield a single method that was effective across all images.

To address this challenge, adaptive contrast enhancement methods could be explored. These techniques dynamically adjust contrast based on the specific characteristics of each image, ensuring consistent and accurate preprocessing across the dataset.

3.5 Conclusion and Future Work

Further research into preprocessing techniques could enhance the accuracy and reliability of the results by tailoring these approaches to the dataset more effectively. Additionally, incorporating object detection using You Only Look Once (YOLO) could improve dataset quality by isolating the plant of interest and removing background elements. This refinement would enhance training reliability and contribute to more robust model performance.

In conclusion, the custom-built Convolutional Neural Network (CNN) model developed in this study achieved an accuracy of 97.35% without reliance on pretrained models. This approach significantly reduced both storage requirements and computational complexity, leading to a more efficient process. In contrast, initial experiments with VGG16, along with insights from the literature, highlighted a heavy reliance on pretrained models in previous research. The findings of this study demonstrate the feasibility of constructing a highly accurate, resource-efficient model tailored for plant disease classification.

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